ICE-SKATING INJURIES TO THE HAND

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166 patients with injuries due to ice-skating were seen in the local accident and emergency department over a six-month period; 60% of these involved the upper limb. 28 of the upper limb injuries were closed fractures and 24 were lacerations, almost all to the dorsum of the hand. The use of protective gloves would help to prevent these potentially avoidable injuries. "On-site" first-aid facilities help to reduce the demand on the local accident service.

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Ice-skating is rapidly increasing in popularity. It has been publicised by wide television coverage and the success of British amateurs, so many local authorities now see it as a leisure pursuit to be made available to people of all ages.

Our interest was aroused by the number of patients presenting to our accident and emergency department with hand injuries after a new ice-rink opened in our area. A small number of these were serious enough to warrant referral to the plastic surgery department. The nature of these injuries is reviewed and their possible prevention discussed.

Material and methods

The case-notes were reviewed of all patients seen at Wexham Park Hospital in the first six months of 1987 with injuries sustained whilst ice-skating. The records from the first-aid station at the ice-rink for the same period were also examined.

A total of 166 such patients was seen at the hospital during this period, of whom 99 (60%) had upper limb injuries. Of these 99, 28 had sustained closed fractures. The sites of fracture are shown in Table 1.

The hand was lacerated in 24 patients, with a total of 30 wounds: of these, 28 were on the dorsal aspect of the hand. Most of these lacerations were "skate-over" injuries caused by the skate blade of a following skater after the patient had fallen with hand(s) outstretched on the ice. Four patients had compound joint injuries with disruption of the extensor apparatus and a fifth had a phalangeal fracture. These five patients were referred for plastic surgical care, and four were admitted: brief details of each are presented at the end of this section.

The remainder had less serious soft-tissue injuries. Injuries were more common at weekends and mainly affected teenagers. They were sustained equally by both sexes in all age groups, except the twenties which had a male preponderance and the thirties which had a female predominance (Fig. 1).

Table 1—Closed fractures of the upper limb resulting from ice skating

Clavicle	2
Humerus	1
Elbow (details unspecified)	1
Radius and ulna	2
Epiphyseal wrist	3
Adult wrist (details unspecified)	1
Colles'	15
Scaphoid	2
Bennett's	1
Total	28

Fig. 1 Age and sex of patients with upper limb injuries.

The first-aid station at the ice rink was staffed much of the time by qualified nurses. They treated many minor soft tissue injuries and referred the more significant ones to hospital. However, many patients came directly to the A. & E. department without visiting the first-aid station (Fig. 2). Figure 2 also shows the overall numbers of patients seen in the A. & E. department over the same period, and the numbers presenting with injuries due to other sports.

Details of the five patients referred for plastic surgical care are as follows:

Case 1: Skate-over laceration involving the D.I.P. joint of the left middle finger, entering the joint cavity and resulting in a mallet finger deformity. Wound toilet, repair and Kirschner wire transfixion, with antibiotic cover. Good result.

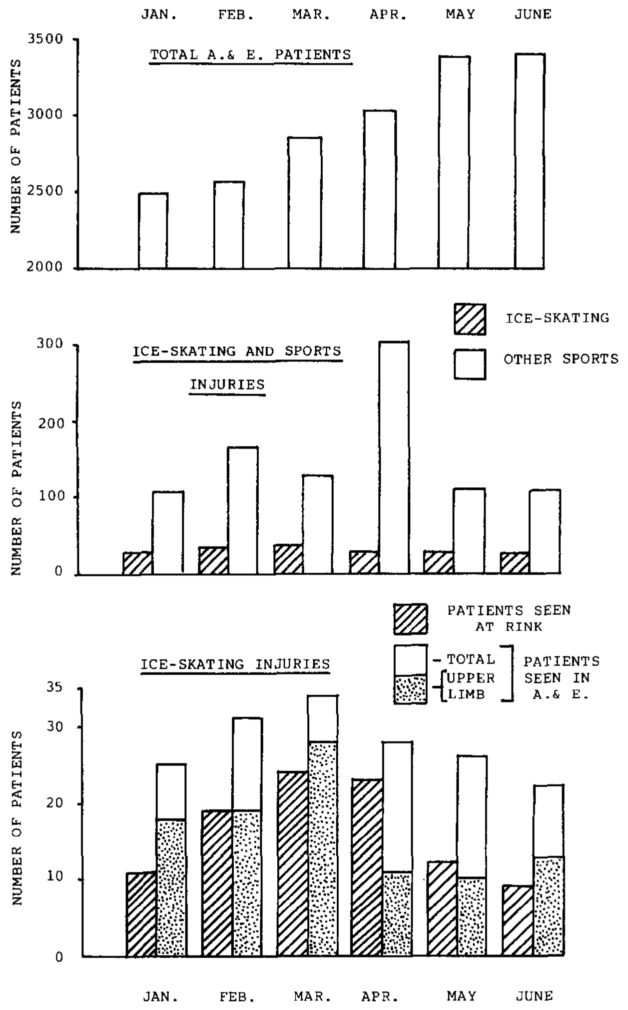

Fig. 2 Histograms showing monthly figures for: (a) Total numbers of patients seen in A. & E. dept. (b) Numbers of patients seen in A. & E. dept. with injuries due to ice-skating, compared to injuries from other sports. (c) Numbers of patients with ice-skating injuries seen at the first-aid station at the rink, compared to numbers seen in A. & E. dept.

Case 2: Skate-over injury of right little finger, causing partial amputation at the level of the D.I.P. joint. Distal circulation adequate. X-ray confirmed compound fractures of both middle and distal phalanges. Fixation with K wire and soft tissue repair. Finger-tip survived, but function poor.

Case 3: Skate-over injury of left index finger. X-ray confirmed an oblique fracture of the middle phalanx which was treated by "neighbour strapping". Patient lost to follow-up.

Case 4: Skate-over injury of right little finger at level of D.I.P. joint causing a compound mallet deformity. The joint was stabilised by K wire fixation, with antibiotic cover. Good result.

Case 5: Skate-over injury of right index finger, dividing the extensor apparatus and exposing the joint space of the D.I.P. joint. A similar injury over the M.P. joint of the thumb did not enter the joint space. Repair and antibiotic cover. Good long-term result.

Discussion

Ice-skating is an expanding leisure pursuit in this country, but until recently there has been little documentation of the possible risks or of the type of injury that may be incurred (Kraus and Conroy, 1984).

Recent reports have shown that the opening of an ice rink can add a significant burden of work on the local accident service (Williamson and Lowdon, 1986; Prescott, 1986; Freeling, 1988). In comparison with these reports, we were not inundated to the same extent. Review of the numbers of patients treated at the first-aid station of the Slough ice rink suggests that treatment there saved many patients with minor injuries a visit to hospital. The ice rink opened in October 1986, but it was several months before the first-aid station became fully effective.

Radford et al. (1988) have reviewed the incidence of ice-skating injuries seen at Oxford in the three years after the opening of a local ice-rink. They noted a definite fall in the number of injured patients seen in hospital over this period and attribute this to improved safety measures at the rink, as well as many minor injuries being treated by the first-aid station at the rink.

Injuries of the upper limb accounted for over half of the ice-skating injuries presenting to our hospital. The majority of these were uncomplicated soft-tissue injuries, without damage to deeper structures; this is in agreement with the findings of Horner and McCabe (1984) and of Radford et al. (1988).

Most of the closed fractures (Table 1) resulted from falls onto the upper limb on the ice. Freeland (1988) found no correlation between the numbers of people attending a rink and the proportion being injured, and Radford et al. found that the majority of those injured in their study had fallen without any collision with another skater. The latter authors did find, as one might expect, that the frequency of serious injuries was much higher in those who were beginners and comment that it is difficult to suggest any simple means of preventing such injuries.

The most commonly-used skates have a metal blade approximately 6 mm wide, with a hollow-ground lower surface so that they have two quite sharp edges. The more serious hand injuries which required referral for plastic surgery care were all due to "skate-over" injuries of the dorsum of an outstretched hand after a fall. In these circumstances, the skate edge cuts soft tissues readily. Surprisingly, most of these injuries did not result in serious damage. Only four of the 24 patients with lacerations were admitted. However, injury to deeper structures is possible, particularly where a laceration crosses a joint, and must be excluded.

In contrast with our findings, a review of ice-hockey injuries by Müller and Biener (1973) showed a lower incidence of hand injuries (7%) and of injuries from skates. This may be explained by the use of protective gloves in that sport.

Conclusions

Our findings suggest the following:

1. The use of protective gloves would prevent many of the hand injuries.
2. With exception of lacerations that cross a joint, the majority of "skate-over" injuries require only simple wound care.
3. The provision of a first aid service "on site" by trained nursing staff reduces the demands placed on the local accident and emergency service.

Acknowledgement

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References

FREELAND, P. (1988). Implications of two newly opened ice rinks on an accident and emergency department. British Medical Journal, 296: 96.

HORNER, C. and McCABE, M. J. (1984). Ice-skating and roller disco injuries in Dublin. British Journal of Sports Medicine, 18: 3: 207–211.

KRAUS, J. F. and CONROY, C. (1984). Mortality and morbidity from injuries in sports and recreation. Annual Review of Public Health, 5: 163–192.

MÜLLER, P. V. and BIENER, K. (1973). Eishockeysportunfälle. Münchener medizinische Wochenschrift, 115: 13: 564–567.

PRESCOTT, M. V. (1986). The effect of opening an ice-rink on the accident and emergency department of a district general hospital. Archives of Emergency Medicine, 3: 107–110.

RADFORD, P. J., WILLIAMSON, D. M. and LOWDON, I. M. R. (1988). The risks of injury in public ice-skating. British Journal of Sports Medicine, 22: 78–80.

WILLIAMSON, D. M. and LOWDON, I. M. R. (1986). Ice-skating injuries. Injury, 17: 205–207.

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